2.5 Proving Statements about Segments and Angles

Name _____ Period ____ Date _____

Geometry 2.5 (30-minute plan) · Student sheet

I can…

- write a two-column proof.
- name and use the properties of segment and angle congruence.
- complete a proof by supplying statements and reasons.

Vocabulary · fill in the missing word

Proof — a logical argument that uses _____ reasoning to show that a statement is true

Two-column proof — a proof with numbered _____ and their reasons in logical order

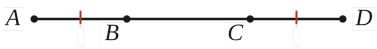
Theorem — a _____ that can be proven

Part 1 · Notes and examples (fill in each note, then work the example)

Two-column proof: every statement needs a _____: given information, a definition, a postulate, a property, or a theorem.

Example 1

Given: A, B, C, D are collinear in that order, and AB = CD. Prove: AC = BD.



Properties of segment congruence: Reflexive ($\overline{AB} \cong \overline{AB}$), Symmetric (if $\overline{AB} \cong \overline{CD}$, then $\overline{CD} \cong \overline{AB}$), and _____ (if $\overline{AB} \cong \overline{CD}$ and $\overline{CD} \cong \overline{EF}$, then $\overline{AB} \cong \overline{EF}$).

Properties of angle congruence work the same way for _____.

Example 2

Name the property.

a. If $\angle 1 \cong \angle 2$ and $\angle 2 \cong \angle 3$, then $\angle 1 \cong \angle 3$.

b. $\overline{PQ} \cong \overline{PQ}$

c. If $\angle A \cong \angle B$, then $\angle B \cong \angle A$.

Example 3

Given: m∠ABC = 90°, D is in the interior of ∠ABC, m∠ABD = 3x°, m∠DBC = (2x + 10)°. Prove: x = 16.

Part 2 · Partner practice

Partner practice

Given: L, M, N, P are collinear in that order, and LM = NP. Prove: LN = MP.